

B. Sc. Semester II (Mid Term Test) 2025**Subject: Statistics****Paper Code: STAMN21**Date: 23rd April 2025

Timing: 9:30 AM – 10:30 AM

Max. Marks: 20

Full Time: 1 Hour

Section- A*Attempt all questions. Each question carries 2 marks.*

1. (i) If A, B and C are any three mutually exclusive and exhaustive events of a random experiment such that $P(B) = \frac{3}{2}P(A)$ and $P(C) = \frac{1}{2}P(B)$. Then find the value of $P(A)$. [2]
- (ii) If A and B are disjoint events of a random experiment, then prove that $P(A) \leq P(A \cup B)$. [2]
- (iii) State Boole's inequality. [2]
- (iv) Define conditional probability and distinguish it from the unconditional probability. [2]
- (v) If A and B are any two mutually exclusive events with positive probabilities, then prove that they are always dependent events. [2]

Section- B*Attempt any two questions out of the following four questions. Each question carries 5 marks.*

2. (i) Prove that the result $P(A \cap B) = P(A)P(B)$ is a necessary and sufficient condition for any two events A and B of a random experiment with positive probabilities to be their independence. [5]
- (ii) Define pairwise independence and mutual independence of any n events of a random experiment. Also prove that for mutual independence of n events $A_1, A_2, \dots, A_n$, we have to check $2^n - n - 1$ conditions. [5]
- (iii) State Bayes theorem. Further, the probability that a doctor H will diagnose a disease D incorrectly is 40%. The chance that a patient will die by his treatment after correct diagnosis is 40% and the chance of death by wrong diagnosis is 70%. A patient of doctor H, who had disease D, died. What is the probability that his disease was diagnosed correctly. [5]
- (iv) Define probability density function (PDF) of a continuous random variable. Next, a continuous random variable X has the PDF $f(x) = 3x^2; 0 \leq x \leq 1$. Find the constants a and b such that $P(X \leq a) = P(X > a)$ and $P(X > b) = 0.05$. [5]

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